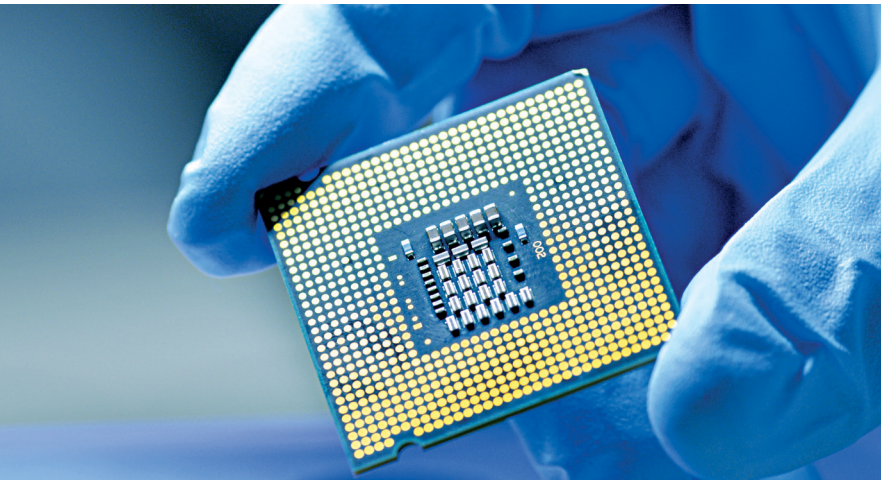


Silicon Ambitions

As demand for chips sustains globally, India is taking small steps towards developing a semiconductor ecosystem

BY NIDHI SINGAL



► **FOR DECADES, INDIA** has dreamt of becoming a player in the semiconductor space, which is currently facing a crippling global shortfall. With the demand for semiconductors expected to only go up, the country needs to develop a comprehensive ecosystem soon. While the government is still taking time to approve proposals to set up silicon fabrication units that produce wafers for chips, some progress has been made on another aspect of manufacturing—ATMP or assembly, testing, marking, and packaging. ATMP units play a key role in slicing the wafers into chips, packaging and testing them—after which they can be used in electronics.

“ATMP is an excellent way to kick-start semiconductor manufacturing in India. It is a relatively low investment compared to wafer fabs and the time to put up the factory is shorter. Depending on the complexity of the package type, the investments can range from \$30-40 million to \$300-400 million,” says Satya Gupta, President of the VLSI Society of India, adding that most global experts suggest that India build ATMP units first. Also, unlike wafers, the output of

ATMP units can be directly consumed by the electronics product companies.

Currently, India has limited capabilities in this domain (or the allied OSAT or outsourced semiconductor assembly and test domain) with one of the few companies doing this being Chennai-based SPEL Semiconductor that does limited types of packages. But with the government’s ₹76,000-crore incentive scheme for the entire semiconductor ecosystem—that has received applications for silicon and display fabs—players are aggressively looking at ATMPs as well. More than half a dozen firms, including Tata Electronics and the Sahasra group, have shown interest in putting up ATMP units in India. Sahasra Semiconductors, for instance, is investing ₹150 crore to set up its ATMP facility in Rajasthan’s Bhiwadi, which it hopes will be operational by the end of the year. Later, it plans to invest an additional ₹600 crore.

Within packaging, there is traditional and advanced packaging. Traditionally each chip area is diced out from the wafer and packaged, but in advanced mode, the packaging is done at the wafer level itself, say industry experts. And it is advanced packaging that Tata Electronics intends to foray into. The company is in talks with some of the large global semiconductor firms and OSAT vendors for this.

There is also a possibility that some of the large global memory manufacturing companies such as Micron will consider packaging their memory chips in India, believe experts who track the industry. In fact, Karnataka’s IT Minister Dr Ashwath Narayan C.N. says that the state is in talks with multiple players for ATMP and OSAT facilities.

Although experts peg ATMPs’ contribution to be about 6 per cent of the overall semiconductor industry, it is still a key part of the ecosystem and could help in creating a fab and fabless ecosystem in the country. The industry would certainly hope for that. **BT**

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